

Writing a Technical Paper

INSTRUCTOR Dr. Chutisant Kerdvibulvech

Textbooks

- How to write and publish a scientific paper, by Robert A. Day, Barbara Gastel, Cambridge University Press, ISBN 0521671671, (2006)
- Proceedings of the International Conference on Artificial Reality and Telexistence (ICAT'09), (2009)

Writing and publishing a research article

Resources

Day, RA. *“How to write and publish a scientific paper,”* 5th edition, Oryx Press.

Thomas H. Adair, “Publishing a Research Article” University of Mississippi

Fischer BA, Zigmond MJ. *“Components of a research article.”*

Marshall GS. *“Writing a peer reviewed article.”*

Hall, JE. *“Writing research papers (and getting them published)”*

Benos, D., Reich, M. *“Peer review and publication in APS journals.”*

*“There is no way to get
experience except through
experience.”*

Why write and publish research papers?

- Ideally –
 - to share research findings and discoveries with the hope of improving healthcare.
- Practically –
 - to get funding
 - to get promoted
 - to get a job
 - to keep your job!

*“Scientists are rated
by what they finish,
not by what they
attempt”*

Getting a paper published

- Competition for space in journals is intense
- Cost of publication is high, \$360/page for APS
- Rejection rates vary
 - AJP = 50%
 - JBC = 65%
 - NEJM, Science, Nature = 90%

Major reasons for rejection

- Confirmatory (not novel)
- Poor experimental design
 - Poor controls
 - Hypothesis not adequately tested
- Inappropriate for journal
- Poorly written

Tips

1. Know the journal, its editors, and why you submitted the paper there
2. Pay close attention to spelling, grammar, and punctuation
3. Make sure references are comprehensive and accurate
4. Avoid careless mistakes
5. Read and conform to “Instructions for Authors”

Publish or perish



"Surely you were aware when you accepted the position, Professor, that it was publish or perish."

Avoid redundant publication

What constitutes redundant publication?

- Data in conference abstract? **No**
- Same data, different journal? **Yes**
- Data on website? **Maybe**
- Data included in review article? **OK if later**
- Expansion of published data set? **Yes**

What makes a good research paper?

- Good science
- Good writing
- Publication in good journals

What constitutes good science?

Novel – new and not resembling something formerly known or used (can be novel but not important)

Mechanistic – testing a hypothesis - determining the fundamental processes involved in or responsible for an action, reaction, or other natural phenomenon

Descriptive – describes how things are but does not test how things work – hypotheses are not tested.

What constitutes a good journal?

Impact factor –
average number of times published papers are cited up to two years after publication.

Immediacy Index –
average number of times published papers are cited during year of publication.

Journal Citation Report, 2003

Journal	Impact Factor	Immediacy Index
Nature	30.979	06.679
Science	29.162	05.589
Hypertens	05.630	00.838
AJ P Heart	03.658	00.675
Physiol Rev	36.831	03.727
Am J Math	00.962	00.122
Ann Math	01.505	00.564

5907 journals

Things to consider before writing

1. Time to write the paper?

- has a significant advancement been made?
- is the hypothesis straightforward?
- did the experiments test the hypothesis?
- are the controls appropriate and sufficient?
- can you describe the study in 1 or 2 minutes?
- can the key message be written in 1 or 2 sentences?

“Those who have the most to say usually say it with the fewest words”

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2. Tables and figures

- must be clear and concise
- should be self-explanatory

3. Read references

- will help in choosing journal
- better insight into possible reviewers

Things to consider before writing

4. Choose journal

- study “instructions to authors”
- think about possible reviewers
- quality of journal “impact factor”

5. Tentative title and summary

6. Choose authors

Authorship

GUIDELINES ON AUTHORSHIP

Each author should have participated sufficiently in the work to take public responsibility for the content. This participation must include: (a) conception or design, or analysis and interpretation of data, or both; (b) drafting the article or revising it for critically important intellectual content; and (c) final approval of the version to be published. Participation solely in the collection of data does not justify authorship.

All elements of an article (a, b, and c above) critical to its main conclusions must be attributable to at least one author.

Guidelines on authorship, *International committee of Medical Journal Editors*,
Reprinted by kind permission of the Editor of the British Medical Journal of Sept
14, 1985. *J Clin Pathol* 39: 110, 1986

Writing the manuscript

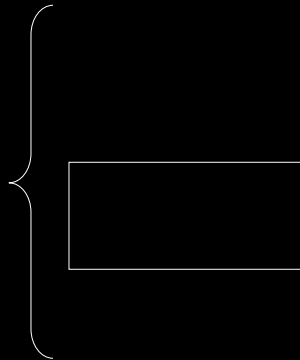
The hardest part is
getting started.

Parts of a manuscript

Title
Abstract
Introduction
Methods
Results
Discussion
Acknowledgements
References

Write in what order?

Title
Abstract
Introduction
Methods
Results
Discussion
Acknowledgements
References



Methods and materials

- Best to begin writing when experiments still in progress.
- Should be detailed enough so results can be repeated by others.
- Reference published methods where appropriate.
- Include animal/human use approval information.
- Use descriptive subheadings
 - Animals
 - Surgical procedures
 - Histochemistry

Results

- Briefly repeating protocols can be effective
- Tables and figures must be straight forward and concise
- Present main findings referring to tables/figures.
- Do not speculate or over discuss results.

Introduction

- Build case for why study is important/necessary
- Provide brief background
- State hypothesis / central question
- Give a one sentence summary of findings

Discussion

- First answer question posed in introduction
- Relate your conclusion to existing knowledge
- Discuss weaknesses and discrepancies
- Explain what is new without exaggerating
- Do not repeat results
- Conclusion/summary, perspectives, implications

References

- Relevant and recent
- Be highly selective
- Read the references
- Do not misquote
- Use correct style for journal

Abstract

- Critical part of paper
- State main objective
- Summarize most important results
- State major conclusions and significance
- Avoid acronyms
- Write and rewrite until flawless

Title

- Will determine whether paper gets read
- Avoid long title (see journal rules)
- Avoid abbreviations
- Title format:
 - “The effects of heat on ice”
 - “Heat melts ice”
 - “The role of heat in melting ice”

Words and expressions to avoid

Jargon	Preferred use
a considerable amount of	much
on account of	because
a number of	several
Referred to as	called
In a number of cases	some
Has the capacity to	can
It is clear that	clearly
It is apparent that	apparently
Employ	use
Fabricate	make

Day, RA. "How to write and publish a scientific paper," 5th edition, Oryx Press, 1998.

Revise, revise and revise

- All authors should participate
- Review order of data presentation
- Polish the writing style
- Double check references
- Look for typos
- Double check spelling

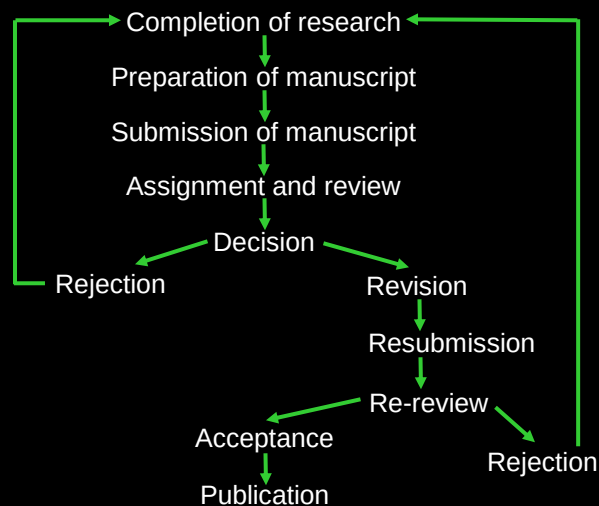
Develop a good writing style

Read well written articles
Try to get good writers to review
Learn from editing changes

Submission

1. Read instructions carefully
2. Fill out all necessary forms
 - Copyright transfer
 - Conflict of interest
3. Write cover letter (suggest reviewers)
4. Confirm receipt after 6 weeks

Process of Research



Responding to reviewers

1. Carefully prepare your responses
 - Each comment should be addressed
 - Each change should be stated
 - Be enthusiastic
2. Reviewer may be wrong
3. Be tactful – thank the reviewers
4. Do not respond to reviewers while upset
5. Never call the editor
6. Get help from other authors

*“There is no way to get
experience except through
experience.”*